

MadhuMarga – Smart Beekeeping Assistant

Android Development Internship Project Report

Submitted By

Name: C S Manoj

USN: 1DA23CS402

Email: 1da23cs402.cs@drait.edu.in

College: Dr. Ambedkar Institute of Technology

Internship Domain: Android Development

1. Project Overview

MadhuMarga is an Android application developed using Kotlin and Jetpack Compose to help beekeepers efficiently manage hive-related activities.

The application helps users:

- Manage hive information
- Track honey production
- Monitor hive health
- Store hive inspection history
- Analyze hive performance
- View seasonal flora guidance

The project demonstrates modern Android development concepts using Jetpack Compose, Room Database, and MVVM architecture.

2. Problem Statement

Traditional beekeeping management is mostly manual and difficult to organize efficiently.

Common problems include:

- Maintaining hive records manually
- Difficulty tracking inspections
- Lack of organized analytics

- Poor monitoring of honey production
- No centralized digital management system

MadhuMarga provides a digital solution to simplify hive management.

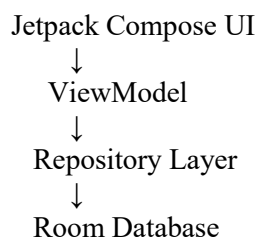
3. Objectives

- Develop a modern Android application using Kotlin
- Simplify hive monitoring and management
- Store hive data locally using Room Database
- Implement analytics and smart hive suggestions
- Create a user-friendly interface using Jetpack Compose

4. Technologies Used

Technology	Purpose
Kotlin	Android development language
Jetpack Compose	UI development
Room Database	Local storage
MVVM Architecture	Code organization
Material 3	UI styling
Android Studio	Development IDE

5. System Architecture



The application follows MVVM Architecture for better maintainability and clean code structure.

6. Features Implemented

Login Screen

- Simple login interface
- Material 3 UI design
- Demo authentication system

Home Dashboard

- Hive information display
- Honey collection tracking
- Queen bee monitoring
- Pest detection status
- Honey flow progress tracking

Add Hive Feature

- Add new hive information
- Store data using Room Database

Hive Inspection History

- Maintain hive inspection records
- View timestamps and inspection status

Flora Calendar

- Seasonal flower guidance
- Helps understand nectar flow periods

Analytics Dashboard

- Total hive count
- Honey production analysis
- Average production statistics
- Best hive performance analysis

Profile Screen

- User information display
- Dynamic hive statistics

Settings Screen

- Notification toggle
- App information section

7. Working Methodology

User Login



Access Dashboard



Manage Hive Data



Store Data in Room Database



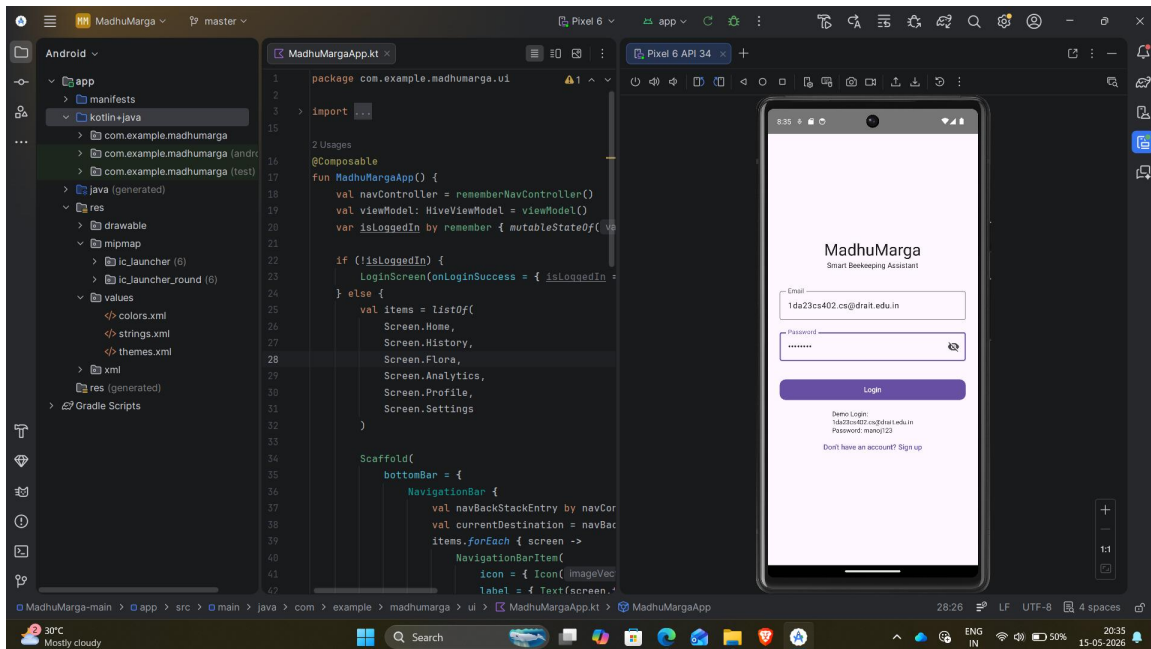
Generate Analytics & Suggestions

The application stores all data locally and updates information dynamically.

8. Application Screenshots

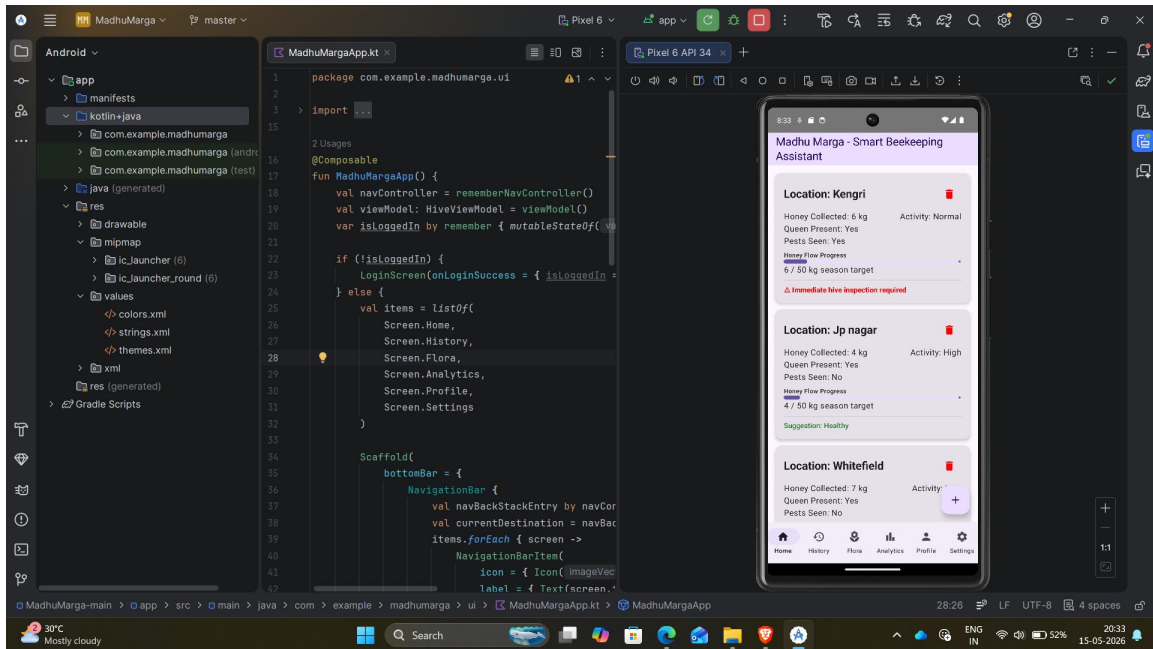
Login Screen

Shows the user login interface developed using Jetpack Compose.



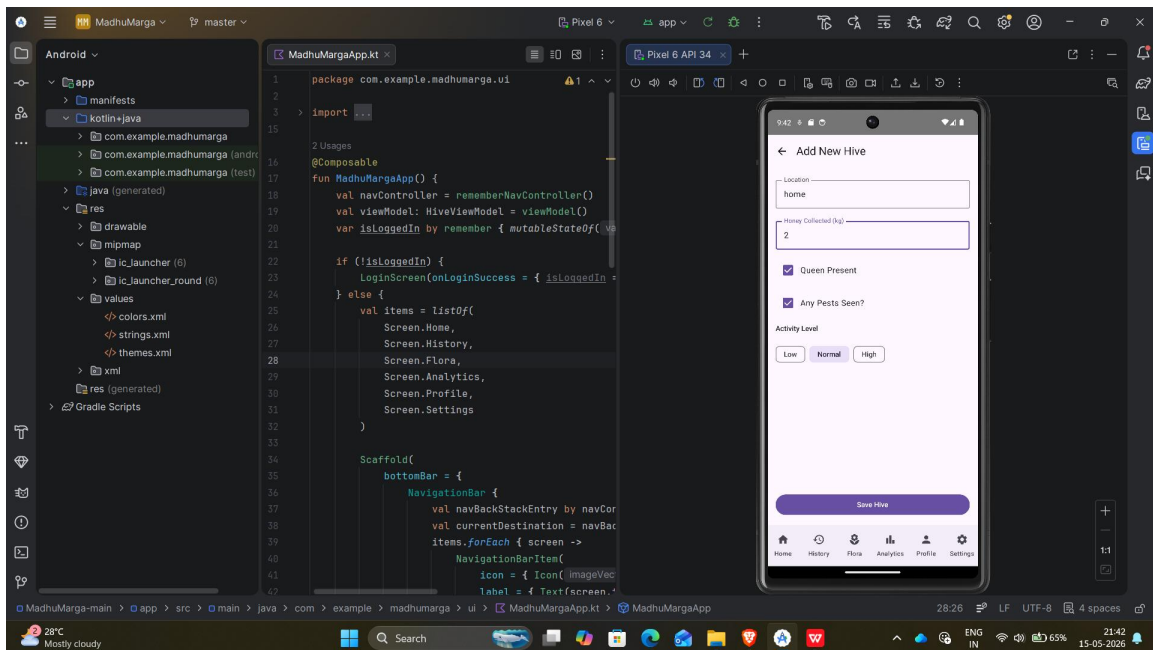
Home Dashboard

Displays hive information, honey collection details, and hive activity.



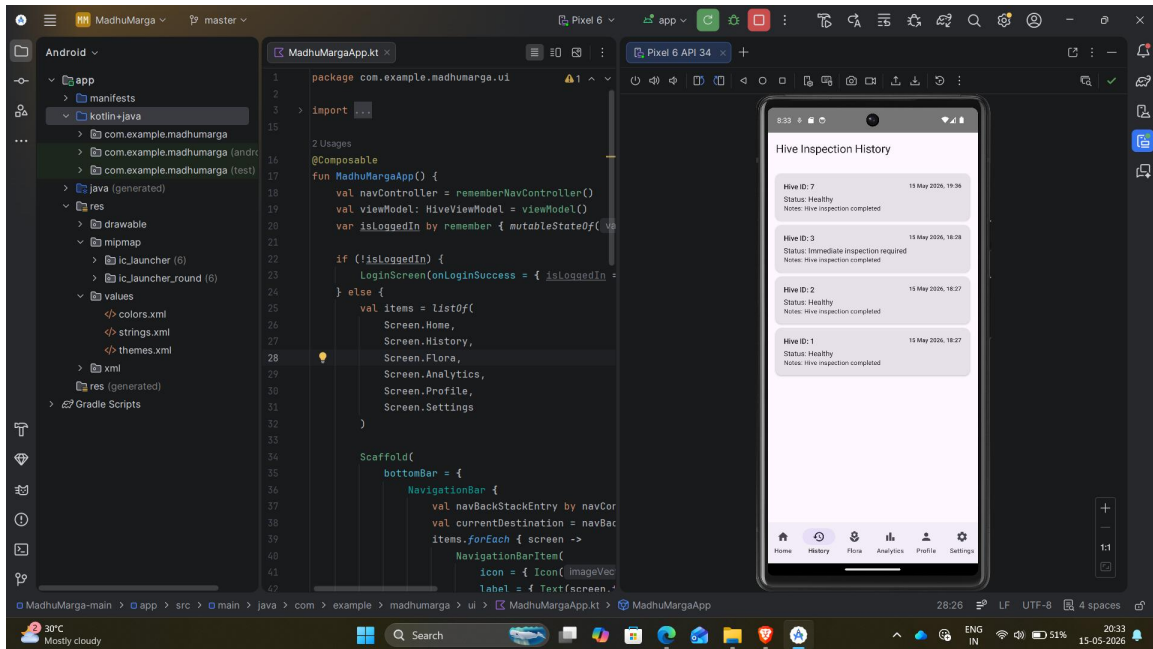
Add Hive Screen

Allows users to add and save new hive details.



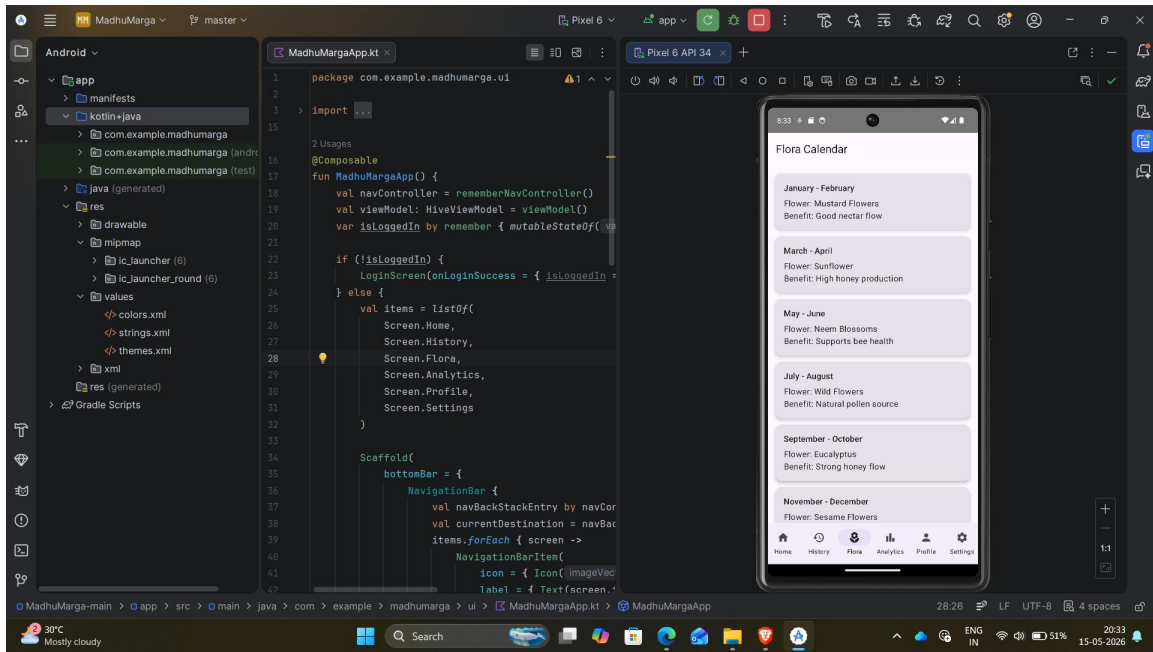
Hive History Screen

Displays inspection records and hive history.



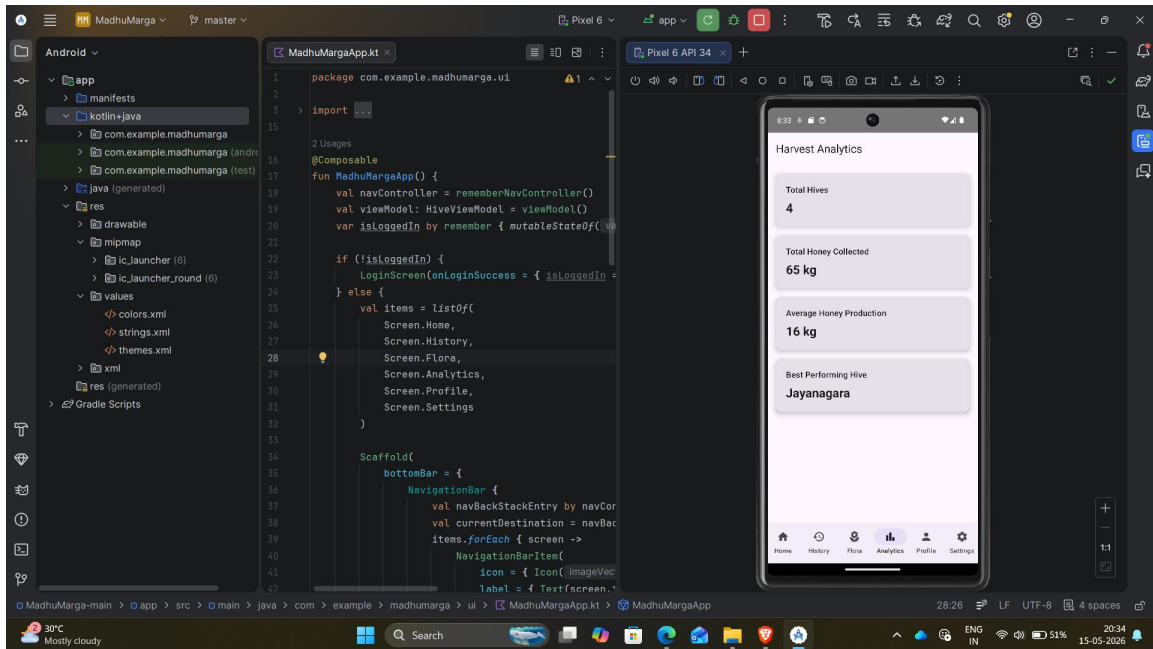
Flora Calendar Screen

Shows seasonal flower guidance useful for beekeeping.



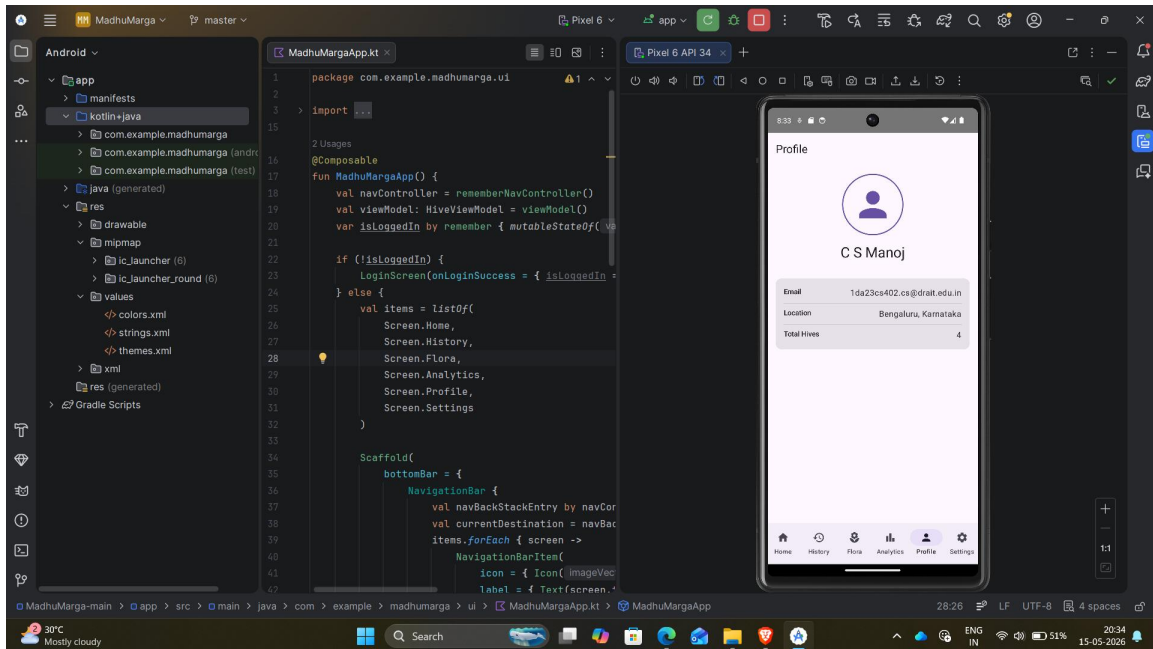
Analytics Dashboard

Displays total hives, honey production, and analytics information.



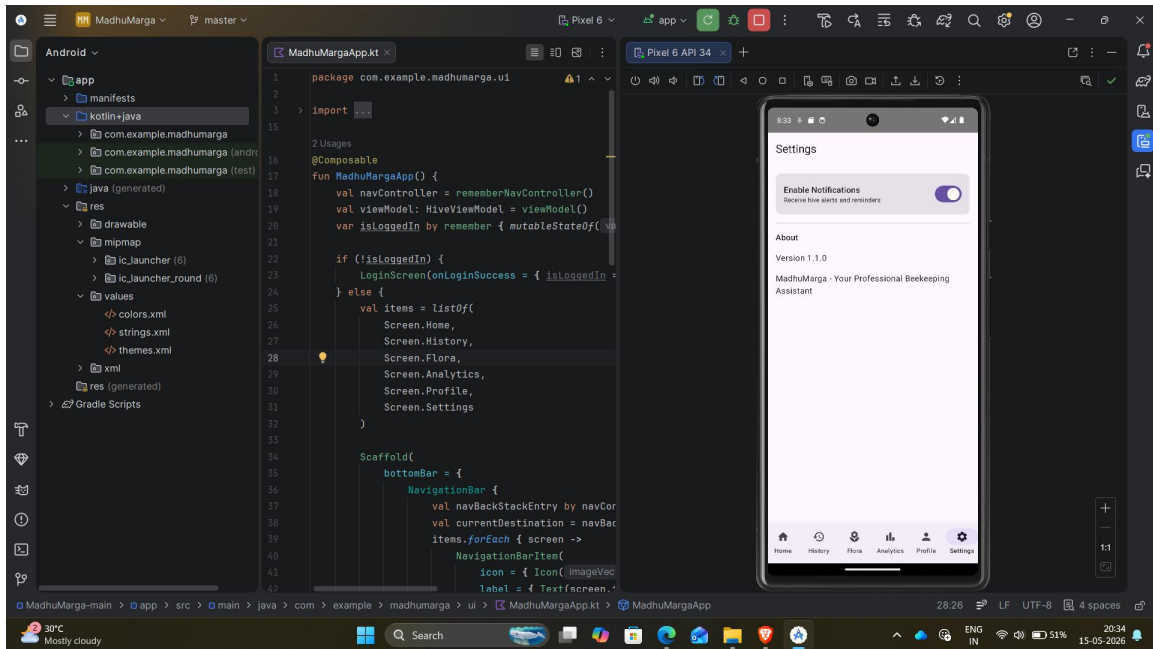
Profile Screen

Displays user information and hive statistics.



Settings Screen

Displays notification toggle and app information.



9. Advantages

- Easy hive management
- Organized data storage
- User-friendly interface
- Offline functionality
- Modern Android architecture implementation
- Better tracking of hive performance

10. Challenges Faced

During development, the following challenges were faced:

- Learning Jetpack Compose layouts
- Managing app navigation
- Integrating Room Database
- Maintaining UI responsiveness
- Managing application state properly

These issues were solved through debugging, testing, and Android documentation.

11. Future Enhancements

- Firebase Authentication
- Cloud data synchronization
- Push notifications
- AI-based hive health prediction
- Weather integration
- Image-based pest detection

12. Conclusion

MadhuMarga successfully demonstrates the implementation of a modern Android application using Kotlin and Jetpack Compose.

The application simplifies hive management through digital tracking, analytics, and local data storage.

This internship project helped improve practical knowledge of Android development, UI design, Room Database integration, and MVVM architecture.

GitHub Repository

<https://github.com/1da23cs402cs-cyber/MadhuMarga>

Developed By

C S Manoj

Android Development Intern

Dr. Ambedkar Institute of Technology